



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF BUILT ENVIRONMENT & SURVEYING
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SBEU 3213 FIELD ASTRONOMY
SECTION EB1

Individual Assignment

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1.0 Conceptual Understanding

1.1 What is apparent solar time?

Apparent solar time is a system of timekeeping based on the significant real movement of the Sun circulating in its orbital eclipses. Solar noon occurs when the Sun is at its highest point and crosses the local meridian. The length of an apparent solar day varies throughout the year due to Earth's tilted axis and elliptical orbit, causing the Sun's apparent speed to change. Because of this irregularity, apparent solar time is mainly used for understanding solar motion and is shown by sundials, while mean solar time is used for precise timekeeping.

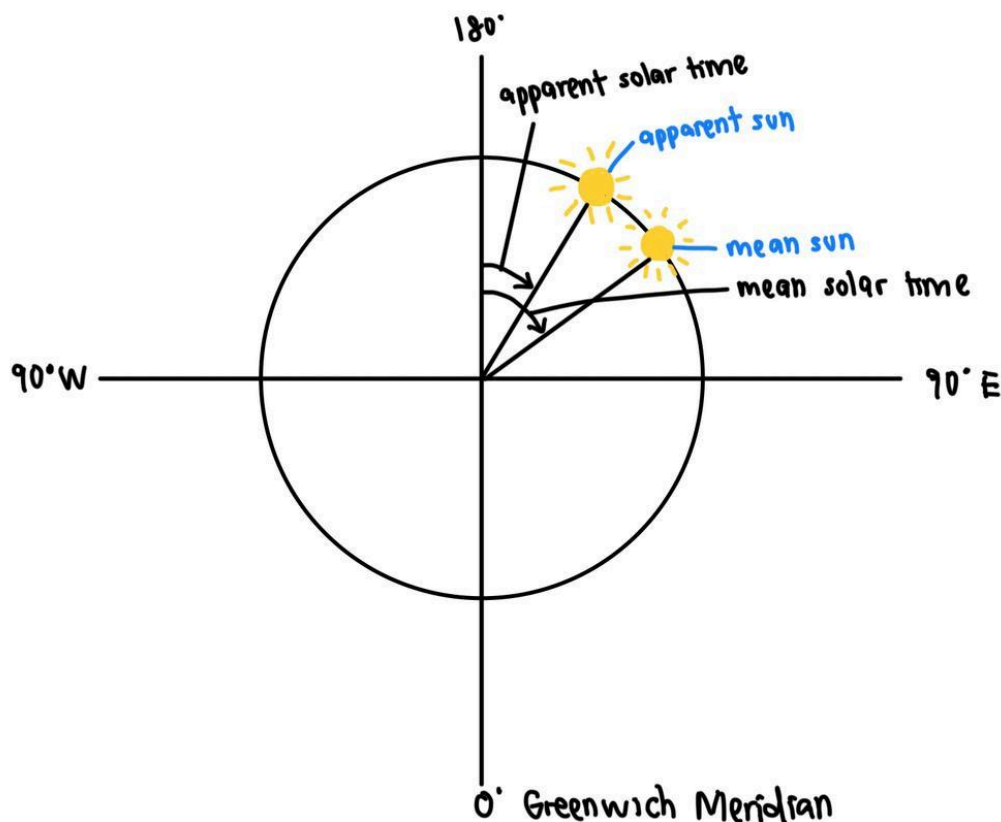


Diagram 1.0 Mean and Apparent Solar Time

1.2 Why must the gnomon be aligned with Earth's rotational axis?

The gnomon must be aligned with Earth's rotational axis so that it remains parallel to the axis about which Earth rotates. When correctly aligned, the shadow cast by the gnomon moves at a constant rate throughout the day, allowing time to be measured uniformly. This alignment

ensures that the sundial indicates time accurately throughout the year, regardless of seasonal changes in the Sun's apparent path caused by Earth's axial tilt.

1.3 How does latitude affect sundial design?

Latitude affects sundial design because the angle between Earth's rotational axis and the horizontal plane changes with location on Earth. To function correctly, the gnomon must be inclined at an angle equal to the local latitude so that it remains parallel to Earth's rotational axis. As latitude increases, the gnomon becomes steeper, and the spacing of the hour lines changes accordingly. Therefore, a sundial designed for one latitude will not accurately indicate time at another latitude unless it is adjusted.

1.4 Why does sundial time differ from clock time?

Sundial time, or apparent solar time, differs from clock time because the length of an apparent solar day is not exactly 24 hours. This variation occurs due to Earth's elliptical orbit and the tilt of its axis, which cause the Sun's apparent speed across the sky to change throughout the year. In contrast, clock time, based on mean solar time, assumes a uniform 24-hour day. Additional differences arise from the sundial's location relative to the standard time zone and the use of daylight saving time.

2.0 Sundial Design

The horizontal sundial was selected for this project and modelled using the software “Shadows Pro”.

2.1 Design sketch

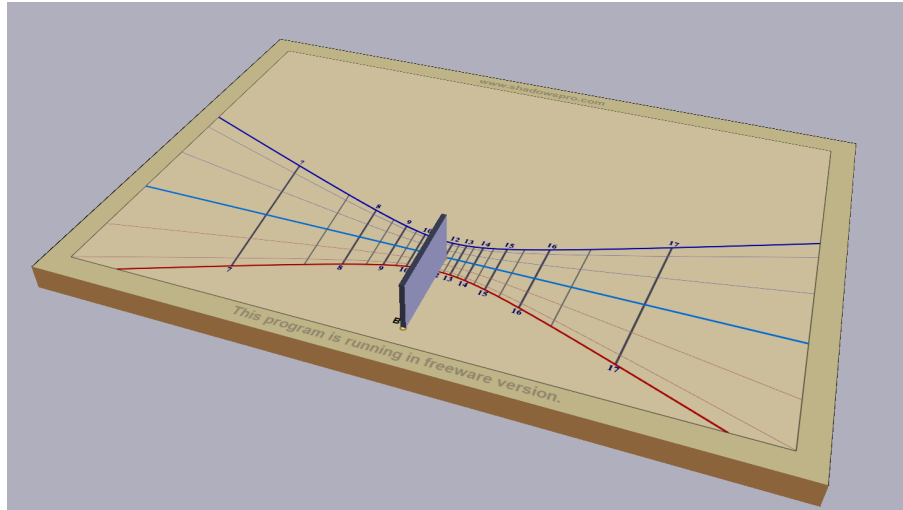


Diagram 2.0 Sundial Design Sketch

2.2 Calculations for gnomon angle and hour lines

2.2.1 Gnomon Angle

$$\text{Gnomon Angle} = \text{Latitude} = 1.576^\circ$$

2.2.2 Hour Line Angles (7:00–17:00 with half hours)

Sundial Time	Angle from Noon Line θ
7:00	5.86°
7:30	3.80°
8:00	2.73°
8:30	2.05°
9:00	1.58°
9:30	1.21°
10:00	0.91°
10:30	0.65°

11:00	0.42°
11:30	0.21°
12:00	0°
12:30	0.21°
13:00	0.42°
13:30	0.65°
14:00	0.91°
14:30	1.21°
15:00	1.58°
15:30	2.05°
16:00	2.73°
16:30	3.80°
17:00	5.86°

3.0 Construction

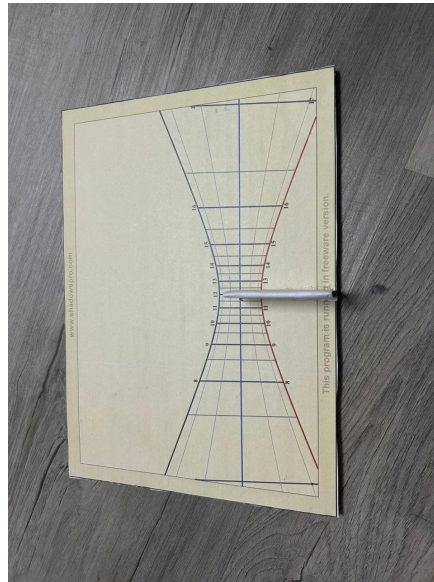
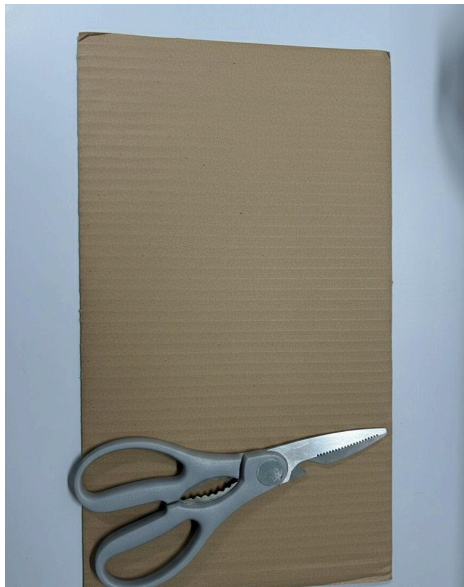


Diagram 3.0 & 3.1 Construction Material and Sundial

The sundial was constructed using cardboard, glue, scissors and a printed sundial surface. The printed sundial surface was attached securely to the cardboard to provide a stable base. The gnomon was then cut out separately and fixed onto the sundial surface, ensuring that it was positioned at a right angle (90°) to the base for accurate shadow casting.

4.0 Field Testing & Observation

Field testing of the constructed sundial was conducted on 17 January 2026 to evaluate its accuracy in measuring apparent solar time. The sundial was placed on a flat, level surface in an open area to minimize shadow obstruction. Gnomon was positioned to face north (0°). During the observation period, the shadow cast by the gnomon was recorded and compared with the corresponding clock time. All readings represent apparent solar time indicated by the sundial.

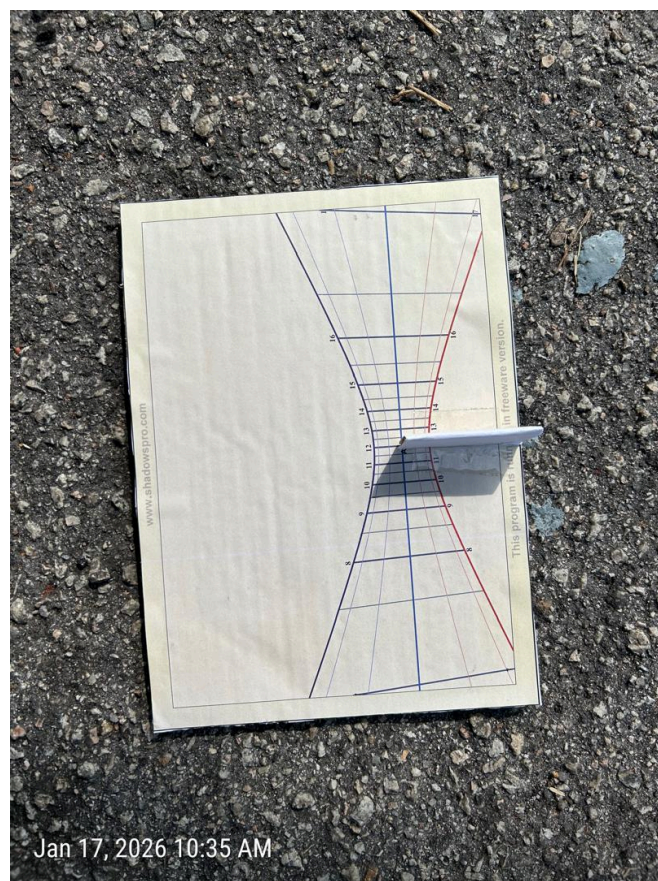


Diagram 4.0 Sundial Reading at 9:30 solar time

4.1 Observation Table

Reading	Sundial Reading	Mean Time (Clock)
1	9:00 a.m.	10:04 a.m.
2	9:30 a.m.	10:35 a.m.
3	10:00 a.m.	11:06 a.m.
4	10:30 a.m.	11:34 a.m.
5	11:00 a.m.	12:07 p.m.
6	11:30 a.m.	12:35 p.m.

5.0 Error Analysis & Evaluation

The sundial readings obtained during field testing were compared with the corresponding local mean time (clock time) to evaluate the accuracy of the instrument. The measured solar time was converted into mean solar time by applying two major corrections, which is the longitude correction and the Equation of Time correction. The corrected values were then compared to the observed clock readings. Discrepancies between the corrected solar time and clock time indicate the combined influence of instrument error and environmental factors. These sources of error are analysed in the following subsections.

5.1 Correction Using Longitude and Equation of Time

5.1.1 Longitude Correction

A sundial indicates apparent solar time, which is the local solar time at the observer's longitude. However, standard time zones are based on a central meridian and apply a uniform time for the entire zone. The longitude correction converts local solar time to the time at the time zone's central meridian using:

$$\Delta t_{\text{long}} = 4 \text{ min/deg} \times (\lambda_{\text{zone}} - \lambda_{\text{local}})$$

where:

λ_{zone} = central longitude of the time zone

λ_{local} = local longitude

For Malaysia (GMT +8), the time zone central longitude is:

$$\lambda_{\text{zone}} = 8 \times 15^\circ = 120^\circ\text{E}$$

Given the location:

$$\lambda_{\text{local}} = 103.620683^\circ\text{E}$$

$$\Delta\lambda = 120^\circ - 103.620683^\circ = 16.379317^\circ$$

Thus the longitude correction is:

$$\Delta t_{\text{long}} = 4 \times 16.379317 = 65.517268 \text{ min}$$

$$\Delta t_{\text{long}} \approx 1 \text{ h } 5 \text{ min } 31 \text{ s}$$

This indicates the sundial is behind the standard clock time and must be advanced by +1 h 5 min 31 s.

5.1.2 Equation of Time Correction

The Equation of Time (EoT) accounts for the variation between apparent solar time and mean solar time due to **Earth's elliptical orbit (orbital eccentricity)** and **axial tilt (obliquity)**.

The EoT value on 17 January 2026 is approximately:

$$\text{EoT} = -10 \text{ min } 2 \text{ s}$$

A negative EoT indicates that the sundial time is slower than mean time, so it must be added to the sundial reading:

$$\Delta t_{\text{EoT}} = +10 \text{ min } 2 \text{ s}$$

5.1.3 Total Correction

The total correction applied to convert sundial time to mean time is:

$$\Delta t_{\text{total}} = \Delta t_{\text{long}} + \Delta t_{\text{EoT}}$$

$$\Delta t_{\text{total}} = 1 \text{ h } 5 \text{ min } 31 \text{ s} + 10 \text{ min } 2 \text{ s}$$

$$\Delta t_{\text{total}} = 1 \text{ h } 15 \text{ min } 33 \text{ s}$$

Therefore, the corrected clock time for each sundial reading is:

Sundial Time	Corrected Time (Mean Time)	Observed Clock Time
9:00 a.m.	10:15:33 a.m.	10:04 a.m.
9:30 a.m.	10:45:33 a.m.	10:35 a.m.
10:00 a.m.	11:15:33 a.m.	11:06 a.m.
10:30 a.m.	11:45:33 a.m.	11:34 a.m.
11:00 a.m.	12:15:33 p.m.	12:07 p.m.
11:30 a.m.	12:45:33 p.m.	12:35 p.m.

The corrected values show that the sundial readings are consistently behind the clock time by approximately 10–12 minutes. This discrepancy is mainly attributed to instrument and environmental errors discussed below.

5.2 Alignment Errors

Alignment error arises when the sundial is not oriented precisely toward true north. A horizontal sundial must have its gnomon aligned with the Earth's rotational axis, i.e., pointing toward the celestial pole. Any angular deviation θ from true north results in a systematic time error approximately proportional to:

$$\Delta t_{\text{align}} \approx (\theta / 15^\circ) \times 60 \text{ min}$$

Even a small misalignment of 1° can produce an error of about 4 minutes. Additionally, if the sundial base is not level, the gnomon's shadow will shift unpredictably, introducing further systematic error during the observation period.

5.3 Gnomon Inclination Errors

The gnomon must be inclined at an angle equal to the latitude of the location (ϕ) to ensure the shadow rotates uniformly with Earth's rotation. The correct gnomon angle is:

$$\theta_{\text{gnomon}} = \varphi$$

At the location:

$$\varphi = 1.575686^\circ \text{ N}$$

Any deviation $\Delta\theta$ causes the hour lines to shift, resulting in time error. Since the sundial is constructed from cardboard, slight deviations in cutting or fixing may occur, especially during morning and afternoon periods when shadow length changes rapidly.

5.4 Atmospheric Refraction

Atmospheric refraction bends sunlight as it enters the atmosphere, causing the Sun to appear slightly higher than its true position. The resulting angular displacement ΔR varies with solar altitude and is most significant near sunrise and sunset. This effect shifts the gnomon shadow and causes a small error in the measured time. While the refraction error is minor during midday, it can still contribute to inaccuracies during the observation period.

6.0 Conclusion

The field study demonstrated that a simple horizontal sundial can provide a reasonable approximation of local time when proper corrections are applied. The sundial readings recorded on 17 January 2026 were consistently behind the mean clock time. After applying the longitude correction and Equation of Time adjustment, the corrected sundial time was calculated to be approximately **1 hour 15 minutes 33 seconds ahead of the raw sundial reading**. The corrected values were still observed to be **10–12 minutes behind the actual clock time**, indicating residual errors in the instrument or experimental setup.

These residual differences can be attributed to several factors, including alignment inaccuracies, incorrect gnomon inclination, and atmospheric refraction. Alignment errors, such as a deviation from true north or an uneven base, can produce systematic time errors of several minutes. Similarly, small deviations in the gnomon angle due to construction limitations can affect the shadow's position, especially during early morning and late morning hours. Atmospheric refraction also contributes to minor discrepancies, particularly when the Sun is at a low altitude.

Overall, the experiment confirms that a cardboard sundial is capable of approximating solar time, but the accuracy is limited by construction quality and environmental effects. For improved precision, the sundial must be carefully aligned to true north, leveled correctly and constructed with higher rigidity. Additionally, using more precise calibration methods and conducting multiple observations can help reduce random errors and increase confidence in the results.